

Summary Of Data Sources And Transformation

- I downloaded three separate datasets from Kaggle, which included Customer details, Product information, and Transaction records from a Business Revenue Analysis dataset.
- Since the original datasets contained a very large number of rows, I opened them in Python to reduce the size for smoother processing in Power BI.
- Using Python slicing, I selected only the first 2,000 rows from each dataset to create a manageable and lighter version for this project.
- After slicing, I saved these filtered datasets as new CSV files so they could be imported easily into Power BI.
- I imported all three CSV files into Power BI Desktop using the **Get Data - Text/CSV** option.
- Inside Power Query, I started the cleaning process by removing all empty or null rows from each dataset to ensure clean and usable data.
- I promoted the first row to headers so that all column names were correctly recognized by Power BI instead of appearing as data.
- I updated the data types for each column (such as text, whole number, decimal, date, and percentage) to make sure all fields had the correct type for calculations.
- I renamed several column names to more clear and readable names so their purpose would be easier to understand later during analysis.
- I also renamed each query to meaningful names like **Customer_Data**, **Product_Data**, and **Transaction_Data** to keep the data model organized.
- After completing the initial cleaning, I started applying different **Text Tools** in Power Query to standardize the textual fields.
- In the **Customer_Data** query, I first used the **UPPER()** text transformation on the *Country_Name* column to convert all country names into uppercase letters for consistency.
- Next, I applied the **LOWER()** transformation on the *Gender* column so that all gender values appear in lowercase format.
- I then applied the **Capitalize Each Word** text option on the *Loyalty_Tier* column to make the category names properly formatted with each word starting with a capital letter.
- After this, I used the **Clean Text** tool to remove any unwanted characters, hidden spaces, or non-printable symbols from the text fields.

- In the **Product_Data** query, I trimmed text in the required columns using the **Trim** tool, which helped remove extra spaces at the beginning or end of the text.
- I also used **Replace Values** to replace the underscore **_** with a hyphen - in the *Brand* column, so values like brand_1 became brand-1 in a cleaner format.
- Finally, I applied rounding to the *Base_Price* column in Product_Data and set it to **two decimal places** to maintain consistent pricing values throughout the dataset.
- After this, I created a new calculated column in the Transaction dataset using **Gross_Revenue** and **Discount_Applied**, with the formula:
Net_Revenue = Gross_Revenue – (Gross_Revenue × Discount_Applied), which gives the revenue after discount.
- In the **Customer_Data** query, I extracted the **Year** from the *SignUp_Date* column to help analyse customer sign-ups across different years.
- In the **Products_Data** query, I applied multiple date extractions on the *Launch_Date* column :-
 - I added a **Month** column.
 - I then added the **Quarter** of each product launch.
 - I also inserted **Day of Week** to understand which weekday products were commonly launched on.
- In the **Transaction_Data** query, I extracted the **Month** from the *Timestamp* column so monthly transaction trends can be analysed more easily.
- After completing the date fields, I created a new column called **Revenue_Status** in the **Transaction_Data** query.
- On this new column, I applied a conditional logic based on **Net_Revenue**:
 - If Net_Revenue $\geq 200 \rightarrow$ *High*
 - If Net_Revenue $\geq 100 \rightarrow$ *Medium*
 - Else $\rightarrow$ *Low*
- I used pivot and unpivot on the Customer_Data table to practice how data can be flipped and rearranged into different formats inside Power Query.
- I grouped the Products_Data table by Product_Category to see all products category-wise and applied aggregation to get useful combined results for each group.
- I performed an aggregation on the Transaction_Data table where I calculated the total revenue for each Product_Id by using the Sum function on the Gross_Revenue column.

- I merged the Transaction_Data and Customer_Data tables using Customer_Id so that customer information gets linked with their transaction records in one combined table.
- I did not perform the Append operation because my tables do not have the same columns, so appending them would not work properly.
- I used the Data Profiling tools (Column Profile, Column Distribution, Column Quality) to check data quality, find missing values, and understand how clean or consistent each column is.
- I worked with Source Settings and Parameters where I set up a dynamic folder path using Parameters and also checked the data source credentials under Data Source Settings.